



Guillaume Falmagne

Research Fellow and team leader in complex systems (TUD and TCD)

Research experience

Education

Research Fellow and Principal Investigator
2025 – 2030 SOHAM centre, TU Dublin & Trinity College Dublin
COOLNET project (€1 million grant):
large-scale cooperation investigated with video games

Postdoctoral research associate
2022 – 2025 High Meadows Environmental Institute, Princeton University
Complex socio-ecological systems
Supervisor: Simon LEVIN

Summer 2024

Complex Systems Summer School
Santa Fe Institute (New Mexico)
4-week interdisciplinary course on complex behavior

Postdoctoral researcher
Subatech (IMT Atlantique), Nantes
Spring 2022 Partonic energy loss in the quark-gluon plasma
Supervisor: François ARLEO

2018 – Dec 2021 The B_c^+ meson in heavy ion collisions with the CMS detector;
Partonic energy loss in the quark-gluon plasma
supervisors: Raphaël GRANIER DE CASSAGNAC, François ARLEO

PhD in particle physics
LLR, École Polytechnique (IP Paris)
Courses: Advanced QFT,
CERN-Fermilab HCP school (2019),
Int. School of QCD (LPT Orsay, 2018),
models for the energy transition, plant seeds

2017–2018 **5-month internship**
feasibility study on PhD topic

MSc (year 2) in High Energy Physics
École Polytechnique (Université Paris-Saclay)
summa cum laude

end 2016 – jun 2017

Volunteering and cultural experience in South America & Eastern Europe

2015 – end 2016 **12-month internship**, CERN, Geneva
 Λ_b^0 production charge asymmetry measurements with the LHCb detector
supervisor: Sascha STAHL

2013–2015 **4-month internship** (2015)
SLAC, Stanford University
NNLO-PS matching & jet clustering in SHERPA
supervisors: Lance DIXON, Stefan HOECHE

**BSc degree & MSc (year 1),
Fundamental Physics**
École Normale Supérieure Cachan (Paris-Saclay)
& Université Pierre et Marie Curie
magna cum laude

2-month internship (2014)
LLR, École Polytechnique
Resolution unfolding of the CMS detector
supervisor: Raphaël GRANIER DE CASSAGNAC

2011–2013

Classes préparatoires in Math & Physics
Lycée Henri Poincaré (Nancy, France)

2011

Baccalauréat in Sciences, summa cum laude

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Awards and fellowships

- 2024 Recipient of the **Prize of the Lopez-Loreta Foundation** for the CORESO/COOLNET project, €1 million grant over 5 years
- 2023 **Best poster (2nd place) of the Collective Intelligence Symposium**, Santa Fe Institute
- 2022 PhD thesis award of **Institut Polytechnique de Paris**
- 2022 PhD thesis award of **Groupeement de Recherche QCD**
- 2022 Accessit to the **Daniel Guinier PhD thesis award of Société Française de Physique**
- 2022 **CERN Senior Research Fellowship**, declined for HMEI Research Associate position, Princeton U.
- 2013 **Admission at ENS Cachan after national competitive exams**, 'normalien': full 4-year funding

Publications

Complex systems

- 2026 **The phase space of symmetric 2×2 games**, Guillaume Falmagne, submitted to *Int. Journal of Game Theory*, short version accepted to *SIAM News*, [SocArXiv tgh9y_v1](#),  Solo author (2025-26)
- 2026 **Inseparability as foundation: rethinking risk science for coupled, co-evolving systems**, Alexandre Dunant + 20 authors, white paper, submitted to *Risk Analysis*
 Contribution to initial IIASA workshop discussions (Nov 2025) and writing (2025-26)
- 2026 **From risk to flourishing: Seeds for responding to a complex world**, Ryan McGranaghan + 8 authors, perspective piece, submitted to *Risk Analysis*
 Contribution to inception, discussions and writing (2025-26)
- 2026 **From phase transitions to tipping points: Theory and empirics of critical transition phenomena**, Elizabeth Krueger + 21 authors, submitted to *Annual Review of Environment and Resources*
 Contribution to inception and Physics review, main *Critical Transitions Workshops* organizer (2023-26)
- 2026 **Interpretable early warnings using machine learning in an online game-experiment**, Guillaume Falmagne, Anna B. Stephenson, Simon A. Levin, **PNAS (2026) 123**, [e2503493122](#),  Lead author (2022-25)

Particle physics (theory)

- 2024 **Path-length dependence of parton and jet energy loss from universal scaling laws**, François Arleo, Guillaume Falmagne, **Proceedings of Science, ICHEP2024 (2025) 604**
- 2024 **Probing the path-length dependence of parton energy loss via scaling properties in heavy ion collisions**, François Arleo, Guillaume Falmagne, **PRD Letters 109 (2024), L051503**
- 2019 **Quenching of hadron spectra in XeXe and PbPb collisions at the LHC**, François Arleo, Guillaume Falmagne, **Proceedings of Science, HardProbes2018 (2019) 075**
- 2016 **Les Houches 2015: Physics at TeV Colliders Standard Model WG Report**, J.R. Andersen et al. (~100 authors), [arXiv:1605.04692](#),  Contribution to NNLO-PS matching in V-Higgs + MINLO procedure

Particle physics (experimental)

Following the publication model of CMS, all papers are signed by all (>2000) authors.

- 2022 **Observation of the B_c^+ Meson in Pb-Pb and pp Collisions at $\sqrt{s_{NN}} = 5.02$ TeV and Measurement of its Nuclear Modification Factor**, CMS collaboration, **PRL 128, 252301**
 Contact author and analysis lead (2018-2021)
- 2021 **Observation of a $\Lambda_b^0 - \bar{\Lambda}_b^0$ production asymmetry in proton-proton collisions at $\sqrt{s} = 7$ and 8 TeV**, LHCb collaboration, **JHEP10 (2021) 060**,  Major role in analysis (2015-17)
- 2019–2021 **Author of all (> 300) papers from the CMS Collaboration submitted in this period**
 Direct contributions to:
Performance of CMS muon reconstruction from proton-proton to heavy ion collisions (**JINST (2024) 19 P09012**),
Fragmentation of jets containing a J/ψ meson in PbPb and pp collisions at $\sqrt{s_{NN}} = 5.02$ TeV (**PLB(2021) 136842**)

Work in progress

Active

- 2025-2026 **Adaptive AI Influence Shapes Human-AI Collective Belief Dynamics**, with Hao Cui and Taha Yasseri (SOHAM, TUD&TCD)
- 2022-2026 **A dynamical model for tipping cascades due to the moisture recycling network in the Amazon rainforest**, with Nico Wunderling (PIK)
- 2024-2026 **Evaluating the effect of network structure on community success using the r/place game-experiment**, with Anna Stephenson (Princeton) and Abraham Israeli (U. of Michigan)
- 2023-2026 **Understanding scaling laws and optimization principles of organizations using the r/place game-experiment**, with Anna Stephenson (Princeton) and Chris Kempes (Santa Fe Institute)
- 2023-2026 **The large-scale propagation of cooperation on multilayer networks**, with Giuseppe Ferro (Princeton), Woi Sok Oh (Princeton), and Emma Zajdela (Princeton)

Paused

- 2024-2025 **Effect of timescales mismatch on large-scale social dilemmas**, with Saverio Perri (UC Riverside), Emma Zajdela (Princeton), Giulio Burgio (UVM)
- 2023-24 **Extracting uni-dimensional voter spectra from ranked choice voting ballots**, with Samuel S. Wang (Princeton), Keena Lipsitz (CUNY), Simon A. Levin (Princeton), and Annie Stephenson (Princeton)
- 2024 **Collective dynamics of singing pitch propagation and convergence**, SFI CSS School project with Galen Hall and Ritam Pal

Scientific talks

- 2025 **Sunbelt**, Social Networks, Paris, *CORES0/COOLNET: new tools for large-scale cooperation*
- 2025 French chapter of the **Complex Systems Society**, Paris, *Interpretable Early Warning System using Machine Learning in a Large-scale Game-experiment*
- 2025 French Conference on Complex Systems, FRCCS, Bordeaux
- 2025 CS2Italy (Computational Social Science), University of Trento
- 2024 "Many Minds" meeting, poster, Dpt of Psychology, Princeton University
- 2024 Workshop on spreading in social networks, DIMACS Rutgers University
- 2024 ERSI Critical Transitions Workshop, Stockholm Resilience Center
- 2024 CoCCoN (Cooperation and Collective Cognition Network) seminar, HU Berlin, online, *CORES0/COOLNET: COOperation in Large-scale NETworks*
- 2024 **European Geosciences Union (EGU) General Assembly**, Vienna (Austria), [online](#), *Interpretable Early Warning System using Machine Learning in a Large-scale Game-experiment*
- 2024 **International School and Conference on Network Science (NetSciX)**, Venice (Italy), *Tipping cascades in the Amazon rainforest due to the moisture recycling network*
- 2023 ERSI Critical Transitions Workshop, PIK, Berlin (Germany), *Trees, networks and games for large-scale cooperation*
- 2023 Seminar at Theoretical Ecology Tea, EEB, Princeton, *Structural aspects of large-scale cooperation*
- 2023 GdR QCD General Assembly, talk for 2022 PhD award, [online](#), *Collective phenomena and critical transitions: from plasma to complex socio-ecological systems*
- 2023 **Collective Behaviour Workshop**, [Isaac Newton Institute](#), Cambridge (UK), *Early warning signals and the structure and emergence of collaborations with a large-scale experiment*
- 2023 Seminar at Max Planck Institute for Human Development, Berlin (Germany), *Early warning signals and the structure of collaborations using a large-scale experiment on Reddit*
- 2023 Seminar at PIK (Potsdam Institute for Climate Impact Research, Germany), *Investigating early warning signals with a large-scale collaborative experiment on Reddit*
- 2023 **Collective Intelligence Symposium**, [poster](#), Santa Fe Institute (NM, USA), *Surveying early warning signals of transitions using a large-scale collaborative experiment*

- 2021 **CERN-LHC Seminar**, [online \(recording available\)](#),
First observation of the B_c^+ meson in PbPb and pp collisions at 5.02 TeV at CMS
- 2021 Rencontres QGP France, [Étretat](#) (France),
First observation of B_c^+ meson production in PbPb and pp collisions with CMS
- 2021 **9th Edition of Large Hadron Collider Physics Conference (LHCP)**, [online \(recording available\)](#),
Exotic quarkonia production in heavy ion collisions: $X(3872)$ and B_c^+
- 2021 **19th International Conference on Strangeness in Quark Matter (SQM)**, [online](#),
First measurement of the B_c^+ meson nuclear modification factor in PbPb collisions with CMS
📖 Published proceedings: [EPJ Web of Conferences 259](#), 12011 (2022)
- 2021 **14th International Workshop on Heavy Quarkonium (QWG)**, [online](#),
Quarkonium production studies in nuclear collisions at CMS
- 2020 Journées CMS-France, [online](#), B_c production: Towards a first observation in heavy ions collisions
- 2019 Journées de Rencontre des Jeunes Chercheurs, [Moulin-Mer](#) (France),
Probing the quark-gluon plasma with the B_c meson in CMS
📖 Published proceedings: C. Armand et al., [JRJC 2019 Book of Proceedings](#), p.88
- 2019 **GdR Intensity Frontier Workshop**, [Sommières](#) (France),
 B_c meson production in pp and PbPb collisions with CMS, and plenary talk (B_c session summary)
- 2019 QGP France, [Étretat](#) (France),
Modification of B_u^+ , B_s^0 and B_c^+ mesons in PbPb collisions with the CMS detector
- 2018 **Int. Conf. on Hard & EM Probes of High-Energy Nuclear Collisions**, [Aix-les-Bains](#) (France),
 B_s^0 and B^+ meson nuclear modification factors in PbPb collisions at 5.02 TeV with CMS detector
📖 Published proceedings: G. Falmagne for the CMS Collaboration, Proc. of Hard Probes 2018, [PoS 143](#)

Teaching experience

- Fall 2024 **Instructor of the *Programming for Biology* course**, EEB department, Princeton University
- 2023 – 2025 Private research mentor, CCIR
- 2018 – 2021 **Teaching assistant**, *École Polytechnique (Institut Polytechnique de Paris)*
Optics, Waves, and Radiation (Bachelor 2nd year). Advanced Particle Physics (Master 1st year).
- 2020 **Research supervisor**, *École Polytechnique*
MSc year 1 (2 months), BSc year 2 (2 months)
- 2014 – 2016 Private tutoring, *LiveMentor*
Mathematics and physics for students in classes préparatoires.
- nov-dec 2014 Teaching internship in physics in high school, *Lycée Frédéric Mistral, Fresnes (France)*

Service

Conference organization

- Spring 2026 **Co-organizer of the SOHAM Seminar**, SOHAM Centre at TUD and TCD, Dublin
- Feb 2025 **Lead organizer of the 6th Critical Transitions Workshop**, including securing funding,
[Princeton Center for Theoretical Science](#), USA
- Dec 2023 **Central organizer of the 4th Critical Transitions Workshop**,
[Potsdam Institute for Climate Impact Research \(PIK\)](#), Potsdam (Germany)

Scientific outreach

- Nov 2023 **Two pictures exhibited in “Birdiversity”**, Princeton
Communicating the science and challenges of bird diversity through art
- oct 2019 Volunteer at **CERN Open Days**, CMS site, Geneva
Outreach on CMS activities, 80,000 visitors in total in a week-end
- apr 2019 Animation of the CERN stand at **Geek Touch Convention**, Lyon (France)
Demonstrating CERN activities to a broad public

Lab and school life

- 2023 Member of the **Climate Committee** of the EEB department
D&I and general climate
- 2019–2022 **Representative** of students and postdocs and sustainable development leader at **LLR Lab Council**
- 2014–2015 Member of the **student association** of the ENS Cachan
In charge of communication and partnerships
- nov 2013 Humanitarian association of ENS Cachan: social microcredit event (*The Rise*, Babylon)

Languages

- French, native/fluent
English
- German Reading, writing, conversation
- Spanish Reading, conversation

Non-professional activities

- Rugby 10+ years in clubs and a university team
- Scuba-diving CMAS two star diver
- Running Typical performance: 10km in 40min
- Music Flute (classical formation, university bands), classical singing (multiple choirs)
- Photography [Travel blog](#), pictures exhibited in “[Birdiversity](#)” (Princeton) and in “[A World Of Stations](#)” (Paris), [instagram](#)